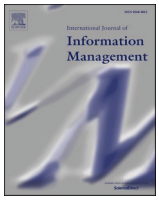
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Opinion Paper



The dynamic effect of interactivity on customer engagement behavior through tie strength: Evidence from live streaming commerce platforms

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ABSTRACT

Interactivity, which is a key characteristic of the live streaming commerce environment, fosters users' active attitudes and behaviors in communications and transactions. However, the literature on live streaming commerce, is scarce, and few studies examine how interactivity influences customers' non-transactional behaviors from a dynamic perspective. In this setting, based on the stimulus-organism-response (S-O-R) framework, we developed a research model using real-time data to investigate the dynamic effect of interactivity on customer engagement behavior through tie strength in live streaming commerce, which is a relatively new derivative of social commerce. This study developed a text mining method to quantify constructs using a large-scale sample of 3,500,445 online review texts. Our empirical study found that interactivity has a curvilinear relationship with customer engagement behavior. Besides, tie strength plays an intermediary role between interactivity and customer engagement behavior. It was further observed that both tenure of membership and popularity have an important moderating relationship between interactivity and tie strength. The study enriches the relationship marketing theory and live streaming commerce literature. Moreover, this study is one of the first studies to use real-time online data for live streaming commerce research.

1. Introduction

Today, Live Streaming Commerce on Social Networking Sites (SNSs), such as Sina Weibo Live and Facebook live, has developed rapidly with the advancement of Internet and ubiquitous usage of mobile devices. Maturing s-commerce, or social commerce, platforms have provided a foundation for the development of live streaming commerce. According to a public report in China, live streaming commerce has achieved a quantum leap forward in development with a year-on-year increase of 400% from 2017 to 2018. Additionally, live streaming content tends to be extremely rich in nature. Consumers watch live streaming for more than 150,000 hours every day and purchase more than 600,000 items on live streaming platforms.[2] Due to the commercial implications of live streaming, many companies, such as Alibaba[3] and Tencent, are calling for academic research on the phenomenon, and our research is a response to those calls.

Live streaming commerce on SNSs typically shows how products are developed and used, presents various facets of products to clarify any areas of customer confusion, and shows live activities that stimulate immediate customer purchases (Lu, Xia, Heo, & Wigdor, 2018). Customers interact in real-time in the live streaming community with broadcasters and other customers. Specially, the broadcasters act as information sources and content contributors, delivering information through customer interactions. Customers as co-creators, participate in the community by writing online reviews. Live streaming contributes to the value of SNSs by supporting online communities (Apiradee & Nuttapol, 2018; Lu et al., 2018), where customers can communicate with each other and co-create value. However, because there are many platforms where online media can replace live streaming for business activities, the conversion cost for customers of live steaming platforms is low, and it is becoming increasingly difficult for the broadcasters to retain consumers. Prior literature on marketing strategies has pointed out that improving community relations and customer engagement is of great help in retaining customers (Apiradee & Nuttapol, 2018; Ponder, Holloway, & Hansen, 2016; Van Doorn et al., 2010). Therefore, it is necessary for the platforms and live-broadcasters to establish a

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[3] http://www.aliresearch.com/ch/information/informationdetails?articleCode=21925.

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harmonious and long-term community relationship based on the interactive features of live streaming commerce, so that they can interact with customers or encourage the interaction between customers in real-time, which make customers to thumb up, generate electronic word-of-mouth (e-WOM), send gifts, etc. These non-transactional behaviors of customers are important manifestations of customer engagement behaviors (Hyeyoon & Jay, 2018; Liu, Shin, & Burns, 2019).

Considering the growing importance of live streaming commerce, we focus on the study of the dynamic mechanism between interactivity and customer engagement behavior through tie strength. As a further exploration in this line of research, the current study is based on the following three cornerstones. The first point is customer engagement behavior. It is a term used in managerial studies to refer to the non-transactional behavior (Van Doorn et al., 2010). However, as a result of the lack of proper technology to obtain data, the behavior intentions rather than the actual behaviors of customers have been quantified in previous studies (Jahn, Kunz, & Meyer, 2013; Jin, 2012). To fill this gap, we employed text mining methods to use actual behaviors (i.e. customers' real-time online reviews) for measuring customer engagement behaviors. Besides, most studies assume that customer engagement is a stable behavior or state, while Brodie, Juric, Ilic, and Hollebeek (2011) suggested that customer engagement may have a dynamic nature with different levels of intensity displayed over time. We followed Brodie, Juric, et al. (2011)'s perspective and investigated customer engagement behaviors from a dynamic perspective.

The second point is interactivity. Although prior research on interactivity has examined the antecedents of customer engagement (Amlia, 2018; Liu et al., 2019; Ul Islam & Rahman, 2017), there is still no consensus on the mutual relationship between customers' engagement behaviors and interactivity. For instance, interactivity affects innovation or new adoption behavior positively (Mark & Rogers, 1997). However, some researchers suggest that interactivity can result in some quitting behaviors like dismissal (Castilla, 2005) or customer churn (Nitzan & Libai, 2011), and may destroy customer engagement. This study significantly differs from the extant literature as we studied the comprehensive impact of interactivity on customer engagement behaviors from a dynamic perspective and also considered its developmental effect. This overcomes the limitation that cross-sectional data can only be used to observe the relationship at a certain time (Kar & Dwivedi, 2020; Liu et al., 2019).

Third, there have been many in-depth studies on interactivity and customer behaviors, but few studies on the internal influence mechanism of interactivity and customer engagement. As a matter of fact, customer engagement behavior is formed spontaneously (Van Doorn et al., 2010) and is a response generated by a certain social stimulation. In general, interactivity is such a social stimulus generated from the external environment. It has also been pointed out by some scholars that the key to produce community prosperity lies in social interaction (Luo, Zhang, Hu, & Wang, 2016), and the internal psychology that drives customers to produce consistent behavior in the community is their innate need to belong (Baumeister & Leary, 1995). Luo et al. (2016) explained that building a harmonious and close virtual community relationship is an important way to satisfy customers' sense of belonging and identification. Tie strength is an important indicator of the closeness and breadth of the relationship among social members. Therefore, based on tie strength, this paper studies the influence mechanism of interactivity on customer behavior under live streaming commerce. And the social ties which may result from repeated interaction between service providers and customers or between customers themselves are highlighted in the current study (Umashankar, Ward, & Dahl, 2017). Besides, social ties have been found affecting referral behaviors in social contacts (Johnson & Reingen, 1987; Ryu & Feick, 2007). In this study, we explore the effect of tie strength as a mediator between interactivity and customer engagement behavior.

This study also discusses the effects of two moderator variables on the relationship between interactivity and tie strength: tenure of membership and popularity. Tenure of membership is a variable because, with the increase of membership length, online community users become more experienced with both the platform and other users (Assmann, Sandner, & Ahrens, 2009), and they learn to adapt themselves to the community thus they can interact more actively with other users (Mittal, Kumar, & Tsiros, 1999). Therefore, we use the tenure of membership to indicate the customer's familiarity with the platform, and familiarity is a critical factor that explains the closeness of social ties (Assmann et al., 2009; Harris, Brookshire, & Chin, 2016). The other potential moderator variable, popularity, is identified as a significant indicator of the quality of products or services (Caminal & Vives, 1996), which in turn is considered as an important factor explaining social ties (Wan, Du, Chiu, & Li, 2018). In the context of live streaming, popularity is reflected as the number of concurrent customers (Chatzopoulou, Sheng, & Faloutsos, 2010; Lin & Kao, 2010)

Based on the above, the fundamental questions that we pose in this study are: (1) What is the dynamic relationship between interactivity and customer engagement behavior in the context of live streaming commerce? (2) How does tie strength, as a mediating variable, work between interactivity and customer engagement behavior? (3) Do factors such as popularity and membership tenure affect the relationship between interactivity and tie strength?

We used the Stimulus-Organism-Response (S-O-R) framework to justify the theoretical background of live streaming commerce. Then, we used a textual similarity approach and tested a conceptual model using a large and real-time data set collected from Sina Weibo, which is akin to a hybrid of Twitter and one of the most popular social network microblogging sites in China. Sina Weibo has an average daily active population of 220 million.[4] Our real-time data set is rendered big data due to its high velocity, huge volume, and a considerable degree of dimensionality. We adopted natural language processing (NLP) to find representative words and improve efficiency in processing text data.

The rest of the paper is organized as follows. In Section 2, we review relevant literature on live streaming commerce, S-O-R framework, interactivity, customer engagement behavior, and tie strength. In Section 3, the main hypotheses of this study will be formulated and proposed. We will then introduce the research design including methodology, the data collection process and variable construction in Section 4. Next, descriptive analysis and empirical results will be presented in Section 5. In the last section, theoretical implications, managerial implications, and future studies will be summarized.

2. Literature review and theoretical background

2.1. Live streaming commerce

As a brand new method of influencing consumer purchase, live streaming commerce is composed of substantial s-commerce and special media attributes (Cai, Wohn, Mittal, & Sureshbabu, 2018; Sun, Shao, Li, Guo, & Nie, 2019). S-commerce is a sub-class of e-commerce, which uses social media that supports interactivity to facilitate online transactions and enrich the online shopping experience of consumers (Jia & Eder, 2011; Liang & Turban, 2011). In this study, we assumed that live streaming commerce has s-commerce attributes that integrate real-time interactivity with e-commerce. Additionally, live streaming commerce, as a new social media form, inherits traditional social media attributes and contains some unique features such as simultaneity (Scheibe, Fietkiewicz, & Stock, 2016) and authenticity (Tang, Venolia, & Inkpen, 2016). Simultaneity means that all user activities happen at the same time and authenticity means that it is an unedited and somewhat unpredictable view into one's life that gives users a feeling of authenticity. In fact, it is realized in two ways, that is, live streaming can be embedded

[4] https://tech.sina.com.cn/i/2020-02-26/doc-iimxyqvz6003265.shtml.

into e-commerce (e.g. Amazon Live Style Code, Taobao.com, Sina Weibo, etc.) and vice versa (e.g. Live.me, Livby, etc.). In our study, we consider the former, that is, the embedding of live streaming into e-commerce.

The live streaming commerce platform is essentially an electronic interactive platform (Cai & Wohn, 2019; Sun et al., 2019). The platform provides an environment for enterprises and customers to co-create value (Cai et al., 2018; Hu, Zhang, & Wang, 2017). As co-creators, customers acquire information value, hedonistic value, and social value by watching live video streaming, interacting with broadcasters and other users, and then form an information pool by sending gifts and commenting. The formation of the information pool provides the possibility of customer engagement (Amlia, 2018), which is an important indicator for enterprises to obtain customer potential value (Zhang, Guo, Hu, & Liu, 2017). Previous studies on live streaming commerce mainly focused on motivations and values. Apiradee and Nuttapol (2018) stated about two aspects of live streaming commerce, sociability and commerciality, they found that the utilitarian and hedonic values affect customer trust which in turn affects customer engagement. Cai et al. (2018) found that hedonic motivation and utilitarian motivation are positively related to celebrity-based intention and product-based intention separately. Almost all previous articles on live streaming commerce focused on motivations and values from a personal perspective and few studies focused on technical perspective of the platforms. Scheibe et al. (2016) showed that the main motivations concerning platforms included ease of use, satisfaction with self-presentation, the level of user-perceived boredom, and acceptance by the community in YouNow (a social live streaming service). Therefore, we considered it necessary to analyze live streaming commerce from the technical perspective of the platforms. The most significant characteristics of live streaming platforms are interactivity (Zhou, Zhou, Ding, & Wang, 2019) and timeliness (Apiradee & Nuttapol, 2018).

Research on live streaming commerce is still in the nascent stage, and most studies have conducted surveys to describe the features of live streaming and consumer motivation (Apiradee & Nuttapol, 2018; Cai et al., 2018). This paper extends the analysis by studying a large amount of real-time customer data to explore the dynamic effect of interactivity on customer engagement behavior. Based on the S-O-R framework, we analyzed the changes in the influence of interactivity on customer engagement behavior objectively and accurately.

2.2. Theoretical framework: the stimulus-organism-response framework

According to the S-O-R theory, S stands for material and social stimuli in the external environment; O refers to individual's internal state or organismic response to the S; and R is the behavioral responses to O such as emotion, motivation, attitude, and reasoning (Mehrabian & Russell, 1974). This model has been successfully applied in s-commerce/e-commerce research for studying user behaviors such as website stickiness (Friedrich, Schlauderer, & Overhage, 2019) and customer loyalty (Yuan, Liu, Su, & Zhang, 2020), and for studying contexts such as online brand community (Ul Islam & Rahman, 2017) and traditional live streaming platforms (Hu et al., 2017).

The use of the S-O-R framework as a theoretical foundation for this research is advantageous for two reasons. First, this model has been previously applied in the study of online consumer behavior to investigate how social interactivity leads to customer non-transactional behavior or adoption behavior (Fengzeng & Ying, 2018; Liu et al., 2019). Hence, this framework seemed suitable for examining consumer behaviors in the context of live streaming commerce as well. Second, a live streaming commerce environment differs significantly from a traditional s-commerce because the former is oriented toward real-time social interactions between broadcasters and customers and among customers. The S-O-R framework allows us to capture these unique elements of live streaming commerce and build a dynamic model to reflect how social interactivity influences customer engagement behavior. Third, the motivation of customers to build a harmonious and close community relationships can be examined by the S-O-R model.

2.3. Interactivity as social stimuli

Although interactivity is defined from different perspectives in literature, scholars generally agree that interactivity refers to the degree to which interactions occur in mutual communication between two parties (Bonner, 2010; Lee, 2005). In studies of social interaction, interactivity is conceptualized as a stimulus (Fortin & Dholakia, 2005; Sheng & Joginapelly, 2011). As an important form of social stimulus in online advertising (Fortin & Dholakia, 2005), interactivity can stimulate customers' physiological arousal thus affect their attitudes and purchase intention. In online commerce, interactivity acts as an important atmospheric cue to stimulate consumers' cognitive and emotional states and subsequently affect their behavioral response (Sheng & Joginapelly, 2011).

Many scholars have addressed elements of interactivity in different situations. Bonner (2010) pointed out three distinct aspects of interactivity: bidirectionality (reciprocal communication), participation, and joint problem-solving in a B2B setting. According to the research, these three dimensions are designed to reflect the intensity and richness of the interaction. Regarding the intensity and richness of the interaction of online community, Yang (2008) conducted qualitative research to identify characteristics that contribute to the perception of interactivity in s-commerce and proposed five schemes that portrayed mobile interactivity: two-way communication, active control, synchronicity, richness of content, and connectedness. Cyr, Head, and Ivanov (2009) also introduced certain characteristics to control the interactivity level in online brand community and to study its impact on three schemes: user control, connectedness, and responsiveness. Besides, Lee (2005) studied six different components (including user control, responsiveness, personalization, connectedness, ubiquitous connectivity, and contextual offer) of interactivity. Although we followed Lee (2005)'s perspective, there is some uniqueness in our context of live streaming commerce. On one hand, the focus of interactivity is on the intensity and the richness of interaction between the customer and the broadcaster and among customers. On the other hand, the live streaming commerce platform as a kind of digital interactive platform naturally possesses the attributes of connectedness, user control, and contextual offer (Ramaswamy & Ozcan, 2018; Sun et al., 2019). Hence, we eventually identified two dimensions of interactivity: responsiveness and personalization which represent the intensity and richness of the interaction respectively. Responsiveness represents the rate of response to an earlier information and personalization reflects the degree to which the information or service is tailored to meet a user's needs (Dholakia, Zhao, Dholakia, & Fortin, 2000).

2.4. Tie strength as organism

Social psychological theories suggest that social reasons like making friends or obtaining emotional support (Ridings & David, 2017) or sharing their feeling and ideas (Park, Kee, & Valenzuela, 2009) are the motivation for individuals to join the online community and participate online communication. This is largely due to their "need to belong" (Gangadharbatla, 2008). Existing as a central element of the theoretical framework of social belongingness, tie strength is a type of fundamental human motivation to form and maintain strong and stable interpersonal relationships with others (Baumeister & Leary, 1995). Umashankar et al. (2017) pointed out that social interaction stimulates customers to establish a certain degree of social ties with the service provider. Therefore, this study posits that tie strength (an organism response) between customers and broadcasters as one of the motivating factor for customers will be influenced by interactivity.

A social tie can be seen as a set of social interactions among two or more individuals (Baumeister & Leary, 1995). Granovetter (1973)

proposed the notion of tie strength, which describes the intimacy of interpersonal relationships. The social penetration theory regards a relationship as a symbol of the breadth and depth of intercommunication, which develops over time, moving from superficial, non-intimate stages to more intimate, deeper stages (Taylor, 1968). Some researchers divide social ties into strong ties (e.g., relative, friend, or neighbors) and weak ties (e.g., acquaintance or stranger) (Chen & Berger, 2013; Dong & Wang, 2018; Johnson & Reingen, 1987), but this method cannot reflect changes in the strength of social ties over time. Therefore, Ponder et al. (2016) framed tie strength as a dynamic process whereby a service provider interacts with the users to develop and improve the service offerings to meet their requirements. To measure tie strength in a dynamic environment, Risselada, Verhoef, and Bijmolt (2014) used the volume of communication between two people, relative to the total communication volume of the focal customer, while Dubois, Bonezzi, and De Angelis (2016) used interpersonal closeness (Someone you feel close to/distant from). In this study, we followed Risselada et al. (2014)'s perspective that tie strength can be measured by the volume of the breadth and depth of communication between broadcasters and customers and among customers in a dynamic environment.

2.5. Customer engagement behavior as response

The last component of the S-O-R framework is the response, which refers to the consequences of social interactions on live streaming commerce platforms, in the form of customer engagement behavior. As businesses strive to encourage customer engagement, researchers attempt to understand customer engagement empirically. Research suggests that customer engagement as a psychological state, occurs by virtue of interactive, co-creative customer experiences with a focal agent/object (e.g. brand) in focal brand/service relationships (Jaakkola & Alexander, 2014). However, some research believes that customer engagement is a non-transactional behavior that results from motivation drivers (Brodie, Hollebeek, Juric, & Ilic, 2011; Brodie, Juric, et al., 2011; Calder, Malthouse, & Schaedel, 2009). Van Doorn et al. (2010) defined customer engagement as a customer's behavioral manifestation toward a brand or a company, beyond the purchase of the product. In the area of s-commerce, customer engagement behavior is usually manifested as the spontaneous behavior in online communities (e.g. eWOM, product referral, thumb up), which may bring value to the company (Dwivedi, Ismagilova, Hughes, Carlson, & Wang, 2020; Zhang et al., 2017). As this study deals with the influence of interactivity on customer behavior in live streaming commerce which is a kind of s-commerce, conceptualization around the connection between customers and broadcasters and among customers as suggested by Van Doorn et al. (2010) was deemed to be the most appropriate choice.

As for measurement, Viglia, Pera, and Bigne (2018) measured customer engagement through visible interactions derived from the cognitive and emotional components of engagement on social media. For instance, customers may engage with firms by commenting on a Facebook page, liking posts on Instagram, retweeting posts on Twitter, or writing a review on Yelp. Therefore, customers' behavioral manifestations occur when a firm's activities on social media stimulate customer engagement (Beckers et al., 2018). Gruss, Kim, and Abrahams (2019) suggested that the numbers of likes, comments, and shares can be viewed as a measurement of customer engagement. Schivinski, Christodoulides, and Dabrowski (2016) considered customer engagement as a holistic concept that incorporates three types of behavioral engagement activities, which are consumption, contribution, and creation. They considered that the customer engagement can be divided into a low level, medium level and strong level. Following this perspective, Liu et al. (2019) identified proxy measures from Twitter data to evaluate the items of consumption, contribution, and creation behaviors. In live streaming commerce, the factor of instantaneity renders the measurement of customer engagement no longer stable and static. Therefore, based on Schivinski et al. (2016) and Liu et al. (2019), we added the measurement of customer engagement under dynamic investigation in live streaming commerce platforms, that is to say, consumption is measured by thumbing up, contribution is measured by gift-giving, and creation is measured by customer reviews.

3. Hypotheses development

3.1. Interactivity and tie strength

We propose that social interactivity is a prominent influential element of tie strength and a basic necessity. Motivated by belongingness, people may have a predominant tendency to build a social connection with each other (Baumeister & Leary, 1995). However, excess interactivity (either through extensive business information or useless information) can lead to stress for customers, thereby reducing customer participation in social networks (Mano & Mesch, 2010; Shen, Huang, Chu, & Liao, 2010). The ever-increasing amount of information in society, worsened by the multiple formats and platforms of communication, results in reduced satisfaction and confidence, and induced negative attitude (Bawden & Robinson, 2009). Hence, increased negative attitudes will hinder customers' willingness to build close relationships with others on network platforms. Then how does interactivity affect tie strength in the context of live streaming commerce? We hypothesize the effects of responsiveness and personalization on tie strength, respectively.

We explore the effect of responsiveness to the strength of social ties firstly. Interactive information technologies' existence is based on conversation (users post messages and wait for a response) (Zhao & Lu, 2012). According to the Social Exchange Theory, when people receive responses from others, they feel fairly rewarded based on the cost or time they invested. Such a balance may persuade customers to build closer relationships in these settings. Besides, Lee (2005) suggested that perceived user responsiveness has a positive impact on a customer's trust in online communities. Once the trust is developed, the psychological distance between users and service providers decreases, and the relationship between them becomes stronger (Brock & Zhou, 2012; Yuan et al., 2020).

Although increased responsiveness to customers' needs can enhance the strength of community relationships, over-responding to customers' demands may cause core rigidity and learning myopia (Leonardbarton, 1992; Levinthal & March, 1993). This can hinder product development and innovative services, and thus can be difficult to attract customers (Zhou et al., 2018).

Concerning live streaming commerce, customers ask questions about product quality or buying channels, expecting real-time, effective responses, securing their relationships with the live streaming community. The more the responses, the closer the community relationships. However, when the number of responses exceeds a certain threshold, the reverse effect is produced. This is precisely because of the lack of information content caused by learning myopia. In other words, under the condition of limited resources, broadcasters may focus predominantly on answering customers' questions, neglecting the creation of unique, high-quality content or services (Sreejesh, Paul, Strong, & Pius, 2020). This leads to boredom from customers, making maintenance of online community relationships difficult (Matthes, Karsay, Schmuck, & Stevic, 2019). Therefore, we suggest the following hypothesis:

Hypothesis 1a. Responsiveness has a nonlinear (inverted U-shaped) relationship with tie strength.

Personalization in social commerce refers that the extent to which content on social media will match a customer's personal preferences and needs (Komiak & Benbasat, 2006). The implementation of personalization strategies is typically in the form of online communication, such as addressing the receiver by their name, or making personalized recommendations based on web data (Tam & Ho, 2006). Just like responsiveness, Lee (2005) also noted that when personalization cues in

online communication are high, users' trust will increase, thus generating the intention to build closer community relationships. More importantly, high personalization causes customers to feel cared for and valued, inducing a keen sense of social identification and social presence (Zhang, Lu, Gupta, & Zhao, 2014). Increased identification and social presence may develop closer personal relationships (Zhang et al., 2014).

However, personalization strategies also come with some psychological costs as they require customers' personal information, such as browsing data (Tucker, 2013), demographic data (White, Zahay, Thorbjornsen, & Shavitt, 2008) or transaction information (Van Doorn & Hoekstra, 2013). This may be extremely personal and may lead to feelings of intrusiveness, causing annoyance and resulting in personal relationship breakdown (Smink, Reijmersdal, Noort, & Neijens, 2020).

Personalization strategies are embedded in real-time interaction in live streaming commerce. Customers may eagerly request information on products or entertainment, in a live streaming room with other customers hoping for a satisfactory reply. Improvements in personalized responses enhance a customer's sense of belonging, promoting the development of the live streaming community. The deeper the personalized communication, the higher the degree of information transparency. However, when personalized responses exceed a certain threshold, customers may perceive a threat to their privacy. For example, some broadcasters use transaction information to recommend specific products repeatedly, which makes customers feel intrusive and not conducive to the development of community relationships. Therefore, we suggest the following hypothesis:

Hypothesis 1b. Personalization has a nonlinear (inverted U-shaped) relationship with tie strength.

3.2. Tenure of membership and popularity

3.2.1. Tenure of membership as a moderating role

Tenure of membership, as a proxy of familiarity, means that the more time users spend on social commerce, the more familiar they become with the platforms and other users. Familiarity is one's understanding of an entity, often based on previous interactions, experience, and learnings of "the what, who, how, and when of what is happening" (Gefen, Straub, & Boudreau, 2000). As familiarity and experience grow, users learn to adapt and interact more frequently with service providers and other users. We argue that the tenure of membership moderates the relationship between interactivity and tie strength.

In general, tenure of membership may induce either trust or distrust, depending on whether the trustor's experience with the trustee is positive or negative (Luhmann et al., 1979). When interactivity is within an appropriate range in the online community, customers experience a keen sense of belonging, bringing positive attitudes toward service providers and other customers (Fred, Lu, Chen, & Farn, 2020). In this situation, experienced customers tend to be more trusting in the community than newer ones. Hence, increased trust will promote the formation of closer community relations. However, excess interactivity can induce negative emotions, such as anxiety and irritation, leading to distrust by experienced customers. Therefore, the tenure of membership reinforces the negative effect of interactivity on tie strength when interactivity is considerably high.

In live streaming commerce, we propose that the tenure of membership amplifies an inverted U-shaped relationship between responsiveness and tie strength. Compared with experienced customers, new customers generally adopt an attitude of having tried (Fred et al., 2020). When responsiveness is low, new customers may think that there is no difference between live streaming commerce platforms and the traditional video-sharing platforms and choose to remain silent, hindering the construction of close community relationships. In this situation, we argue that tenure of membership will amplify the positive effect of responsiveness on tie strength when responsiveness is relatively low. However, experienced customers may devote more attention to information content. When responsiveness is high, the sociality of live streaming community obscures other characteristics of the community (Sreejesh et al., 2020), such as product exhibition, brand shaping and advertising impact. Communities that are unable to meet the needs of experienced customers create distrust and damaged relationships, as these customers have a better understanding of their needs. Therefore, we hypothesize the following:

Hypothesis 2a. Tenure of memberships as a moderating effect will strengthen the inverted U-shaped relationship between responsiveness and tie strength.

Next, we explore the moderating effect of membership tenure between personalization and tie strength in live streaming commerce. Based on the theory of reasoned action, IT adoption, and trust, Komiak and Benbasat (2006) inferred that both personalization and familiarity could effectively increase a customer's trust. When personalization is low, improving customers' familiarity with the platforms increases their trust. Thus, increased trust to broadcasters and other customers may promote tie strength in a community.

However, when personalization is high, we argue that high tenure of membership would dampen tie strength in live streaming commerce. Familiarity reduces expected uncertainty by increasing the understanding of past things (Luhmann et al., 1979), therefore, customers with high tenure of membership are more rational, and privacy protection in the platforms becomes more important. But high personalization often involves the privacy of customers, which may cause a stronger sense of intrusiveness by experienced customers and in turn hinder the building of close community relationships. Hence, we hypothesize the following:

Hypothesis 2b. Tenure of memberships as a moderating effect will strengthen the inverted U-shaped relationship between personalization and tie strength.

3.2.2. Popularity as a moderating role

Prior research has thoroughly explored the importance and complexity of popularity. On the one hand, popularity stands for the quality of a product or service (Pavlou & Gefen, 2004; Zhu & Zhang, 2010). The higher the popularity, the better the quality, and high quality will bring customer trust and satisfaction (Clemons, Gao, & Hitt, 2006). On the other hand, popularity may bring about herding behaviors, where customers follow the group choices regardless of personal needs, product or service quality (Banerjee, 1992; Parthasarathy & Bhattacherjee, 1998). According to herd behavior theory, when the quality of products or services selected by customers through herding behavior is not good enough or cannot meet customer needs, it will arouse strong dissatisfaction, and then destroy the trust and good relationships between customers and company (Josephs, Larrick, Steele, & Nisbett, 1992; Zeelenberg & Beattie, 1997). In response to this paradox, Wan et al. (2018) noted that customers could not choose products by popularity solely, but should be based on whether they can perceive that the product has a certain quality. Therefore, whether the product or service has a certain quality is the key.

Prior literature has showed that interactivity in social commerce is considered as an important indicator of service quality (Barnes & Vidgen, 2002; Parasuraman, Zeithaml, & Malhotra, 2005). When interactivity is low, customers' perceived service quality is equally low. Increased popularity may cause customers to follow blindly, and once service does not meet the requirements, distrust and regret follow. However, when interactivity is high, the perceived service quality increases. Customers who select service due to popularity may associate their choice with building closer relationships.

Popularity is even more important in the live streaming commerce context. In general, the live streaming room is ranked by its popularity, as popular rooms are more noticeable and attract customers further. Therefore, the advantages versus costs of popularity need to be

discussed. When the responsiveness or personalization of the live streaming room is low, the customer's perceived service quality is low accordingly, and high popularity will hinder the formation of close community relationships; when the responsiveness or personalization is high, the customer's perceived service quality is relatively high. Although information overload and privacy issues can lead to the decline of close community relations, the live streaming room with high popularity may slow the rate of decline. Thus, we hypothesize:

Hypothesis 3a. Popularity as a moderating effect will weaken the inverted U-shaped relationship between responsiveness and tie strength.

Hypothesis 3b. Popularity as a moderating effect will weaken the inverted U-shaped relationship between personalization and tie strength.

3.3. Interactivity and customer engagement behavior

We argue that interactivity of live streaming commerce platforms has a considerable impact on consumer engagement. Vries, Gensler, and Leeflang (2012) pointed out that the interactivity of SNSs is important because it promotes customer reactions, such as liking and commenting on a post. Ul Islam and Rahman (2017) stated that interactivity motivates customers to continuously participate in an online community, which positively influences customer engagement. However, paradoxically, information is transmitted through interactions on SNSs, which causes social influence (Nitzan & Libai, 2011). Nitzan and Libai (2011) pointed out that social influence can lead to customer defection. The first reason is that defections usually result in negative WOM, which in turn produces further defections (Wangenheim, 2005). It is generally believed that the effects of negative WOM is stronger than that of positive WOM (Dwivedi et al., 2020; Goldenberg, Sangman, Lehmann, & Hong, 2009; Nam, Manchanda, & Chintagunta, 2010). What's more, individuals are inclined to notice and discuss others' defections, which might help to legitimize planned defections that have not yet taken place.

Responsiveness or response rate is closely related to the degree of exchange of information. According to Tsai and Kang (2019), information exchange in online communities arouses customers' behavior intentions. However, without guidance, users may transmit negative information through the online community, which may lead to customer defections (Chang & Li, 2014). Besides, customers often find it difficult to describe their true demands in online communities because demand knowledge is sticky (Von Hippel, 1994). Hence, some demands may be misleading, and responding to them may cause customers' reactance and avoidance.

In the context of live streaming commerce, we combine the benefits and costs of responsiveness to explore a curvilinear effect on customer engagement behavior. To obtain information for meeting the needs, customers will frequently raise various questions in the live streaming room. With the increase of response frequency, customers may send eWOM or virtual gifts spontaneously. However, if the broadcaster does not excavate the real demands of customers, frequent responses may lead to large-scaled customer churn. In this situation, customer churn would cause an inactive atmosphere in the live streaming room, which is harmful to the content generated by the engaged customers. Therefore, we hypothesize the following:

Hypothesis 4a. Responsiveness has a nonlinear (inverted U-shaped) relationship with customer engagement behavior.

Regarding personalization, Nowak, Shamp, Hollander, and Cameron (1999) revealed that personalized messages increase participant's motivation to attend to marketing-related information. Lacey, Suh, and Morgan (2007) pointed out that personalization is important for promoting a customer's overall commitment to a company. However, personalization strategies may induce customer power, which means customers' claims get more legitimate and influence get more reinforced in online communities (Tsai & Pai, 2012). In this situation, if some customers believe the online community has failed to satisfy their demands, they may feel betrayed and choose to punish a firm by avoidance or sending negative WOM. These negative effects may prohibit customer engagement behaviors.

In the live streaming commerce platforms, increased personalized communications would motivate customers to thumb up, send gifts or eWOM, because consumers prefer showing more positive reactions toward self-relevant information (Kalyanaraman & Sundar, 2006). However, excessive personalized communication gives customers too much power, encouraging them to unite and resist the formation of live streaming communities. For example, customers hinder communication by swiping the screen or booing in live streaming room, deliberately annoying the broadcasters or other customers, leading to the departure of the engaged ones. Therefore, we hypothesize the following:

Hypothesis 4b. Personalization has a nonlinear (inverted U-shaped) relationship with customer engagement behavior.

3.4. Tie strength and customer engagement behavior

Prior research has pointed out that the strength of social ties may influence customers' purchasing intentions. For example, Baker, Donthu, and Kumar (2016) explained that negative eWOM resulting from strong ties would decrease purchasing intentions. Recent literature has increasingly focused on the influence of social ties on customers' non-purchase behaviors as these indicators will assist companies in improving their potential value. Reingen and Kernan (1986) suggested that product referrals can be obtained from stronger ties than weaker ones. Subsequently, Levin and Cross (2004) showed that knowledge received from strong ties is perceived as more useful than that from weak ties. Strong ties also lead to frequent WOM recommendations that are consistent with the needs of the recipient (Dong & Wang, 2018; Koo, 2016; Xu, Guo, Li, Lau, & Liao, 2012). All these non-transactional behaviors are the key to the formation of customer engagement. Thus we have the following hypothesis:

Hypothesis 5. Tie strength has a positive relationship with customer engagement behavior.

Considering Hypothesis H1-H5, the possibility also existed that tie strength may act as a mediating role between interactivity and customer engagement behavior.

A number of studies have explored how interactivity can influence customers' non-transactional behaviors from the perspective of relationships (Amlia, 2018; Guo, Lu, Kuang, & Wang, 2020; Shao & Pan, 2019). However, considering the mediating role of tie strength we do not find empirical evidence in prior literature that sufficiently demonstrates that interactivity will have a curvilinear impact on customer engagement behavior through tie strength in the context of live streaming commerce. Therefore, we hypothesize the following:

Hypothesis 6. Tie strength has a mediating effect on the relationship between interactivity and customer engagement behavior.

4. Research methodology

We developed a conceptual model to examine the dynamic effect among interactivity, tie strength, and customer engagement behavior. The model is two-phased as shown in Fig. 1. In the primary phase, we examined how social interactivity influences tie strength in live streaming commerce platforms. In the second phase, we examined the influence of tie strength and customer engagement behavior.

4.1. Data description

This study was conducted on the live streaming mobile platform Sina Weibo. The copious use of mobile devices is promoting the development

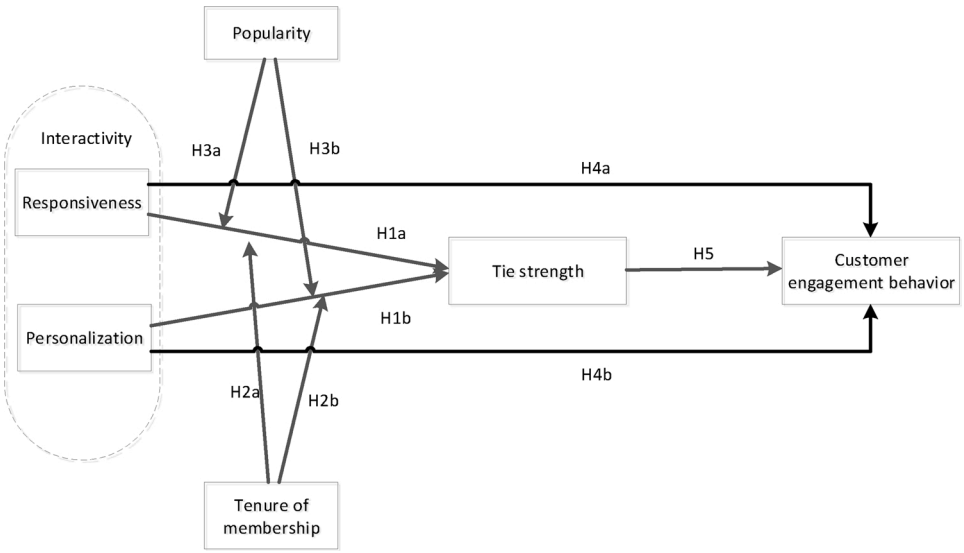


Fig. 1. The research model.

of e-commerce and s-commerce. In live streaming communities, many customers are willing to share and contribute their first-hand demand-side knowledge to assist businesses in improving customer engagement through online reviews (Apiradee & Nuttapol, 2018). Therefore, both the volume and velocity of demand-side knowledge obtained from online customer reviews has grown significantly recently, and it is difficult for managers to efficiently determine customer needs from online big data and consumer remarks.

In this study, we examined our hypotheses with real-time online reviews collected from Weibo live streaming, which is one of China's largest live streaming commerce platforms. Sina Weibo live has become the most popular platform for live streaming commerce in China and was the earliest platform for displaying and selling products in China. Customers are willing to communicate and interact on Sina Weibo live and to reply to other customers' questions. Thus, customers who engage in live streaming have a closer relationship, which is conducive to customer engagement. Many scholars view this platform as an appropriate research context for customer engagement behavior. Since fewer fans mean fewer comments, we chose the number of fans on the top 15 live streams for our research.

The data set we employed in our research is regarded as a typical big data set. We gathered real-time online reviews and sellers' reply text between April 7, 2018, and December 25, 2018, in 15 live streaming rooms and built a panel set for further analysis. The data are divided into the first two months and the last six months, in which the first two months were the sample group and the last six months were the experimental group. Accordingly, we obtained 630,000 online reviews from 52,000 customers during the first two months and obtained 3.5 million online comments from 0.5 million consumers during the next six months. The data provided sufficient indicators including communication details (e.g., about the seller, about the product, and about the response) and comment details (e.g. content, time, volume, etc.). Additionally, we collected data focusing on customers and the live streaming room (e.g., membership tenure, popularity).

The collected data implied three predominant features of big data: (1) high velocity (online consumers write comments in real time), (2) large volumes of information (detailed communication of 15 live streaming rooms; 0.55 million consumers and 4.1 million comments from 0.55 million consumers); (3) a variety of information (unstructured data from varying users and linguistic differences). We deleted emojis and brief comments because they lacked value. Therefore, the final data set included 3,500,445 reviews from 510,345 customers.

4.2. Variable measurement

In this section, we measured variables from real-time online reviews using text mining. The original data were collected from web crawling Sina Weibo Live. Since the unstructured data must be transformed from short phrases in Chinese to structured ones, we used the jiebaR software to perform word segmentation. Besides, due to different backgrounds, professions, and experience of customers, they are likely to use different communication modes for the same subject, which may lead to a deviation in the measurement of variables. Thus, we used a text similarity method, word2vec and SVMperf,[5] to solve the problems of text classification. Word2vec, which is used for the latent semantic analysis, was used to categorize the keywords relating to the identical characteristic. SVMperf was used to categorize different types of reviews according to their features, which are based on keyword similarity in the reviews. The method proposed in this study combines word2vec with SVMperf (Zhang, Hua, Su, & Xu, 2015). Moreover, since we want to explore real-time online reviews, it is necessary to establish sample data to train the text set, so that we can classify the real-time texts according to the text set. We used a framework to reflect data analysis process as shown in Fig. 2.

Interactivity (independent variables: H1a–H4b) and tie strength (dependent variables: H1a–H3b; independent variable: H5) are both

[5] As a deep learning approach, word2vec is typically applied to reveal the semantic structure in a texts clustering. We adopted the continuous bag-of-words model (CBOW), a mainstream word2vec architecture, to discover representative words and set up the linguistic texts by plotting them on a continuous vector space. First, we applied POS tagging. We used the ICTCLAS4 system developed by the Institute of Computing Technology, Chinese Academy of Science, to disassemble the gathered Chinese review texts into words and assign them adequate POS tags. After expurgating pause words and punctuation marks, we obtained a training set. Second, we used word2vec to train the training set to obtain the vector model. The word2vec tool receives a training set as input and then generates a vector model as output. Third, word2vec provides a command (called "distance") to compare the input word with other words in semantic similarity. In other words, word2vec scans all words and their corresponding vectors in the vector model and uses cosine similarity to compute the semantic distance between the input words and other words. The higher the cosine value appears, the closer the two words semantically are. After sorting the values, we obtain a list of the closest words for input words and their distances. Finally, we use SVMperf to categorize different types of reviews according to their features, which are based on keyword similarity in the reviews.

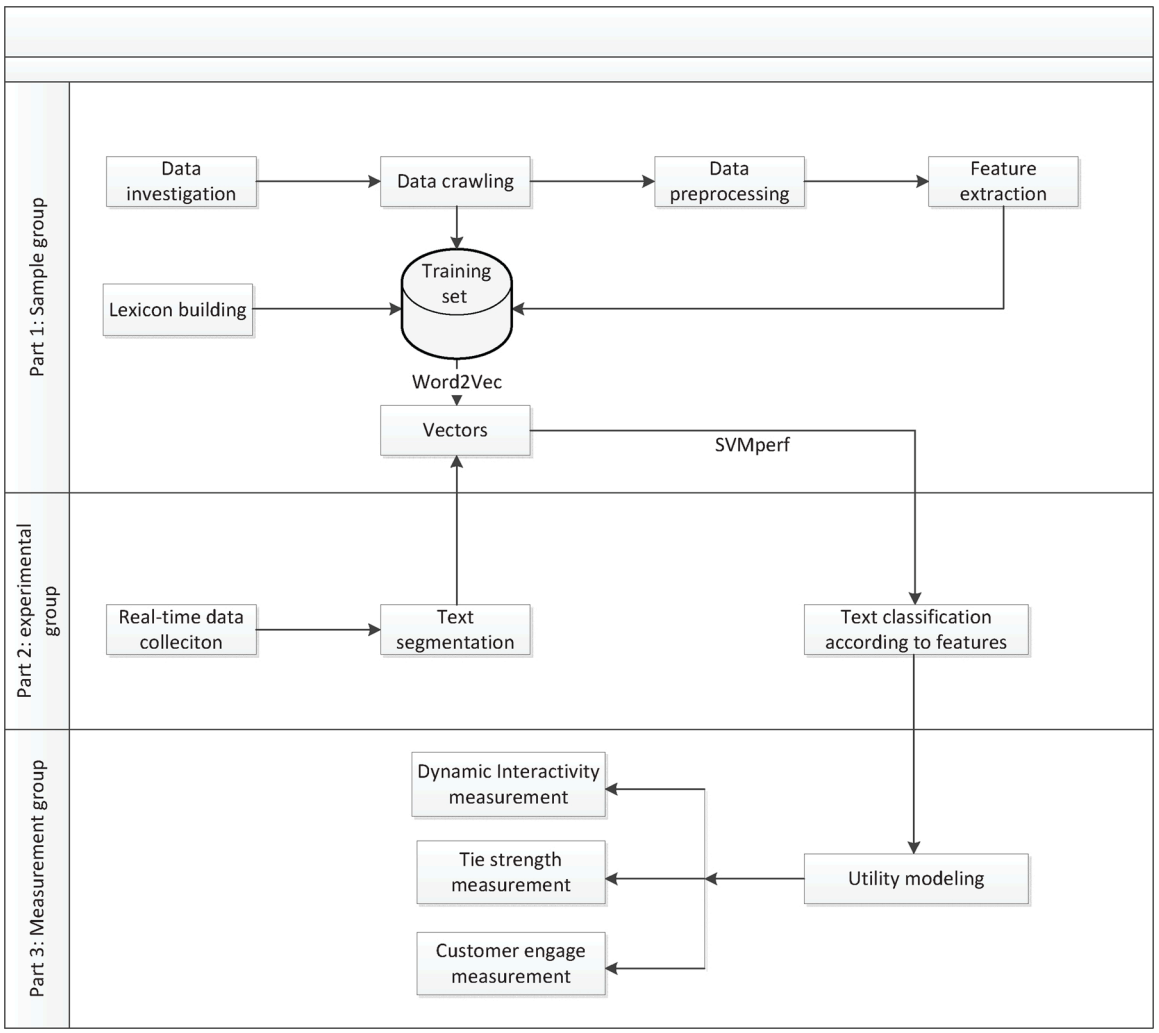


Fig. 2. Research framework.

important constructs of our study. To better reflect the law of changes, we used 10 minutes as a unit of time. We used responsiveness and personalization to reflect the intensity and richness of interactivity respectively. We measured responsiveness as response rate for customers' questions in unit time and measured personalization as average length of response content in unit time. For tie strength, we used the volume of breadth and depth of communication between broadcasters and customers and among customers within a period of time. Specifically, the sum of comments about broadcasters in unit time was seen as the depth of communication and the sum of comments about products in unit time was seen as the breadth of communication. To calculate the composite measures, we chose a commonly used additive aggregation method in which the average of multiple indicator variables was calculated.

Customer engagement behavior (the dependent variable of H4a–H5) is measured by three types of behavioral engagement activities: consumption, contribution, and creation. In the context of live streaming commerce, we added the measurement of customer engagement behavior under dynamic investigation, that is to say, we used the number of liking in unit time as a proxy of consumption, the number of gifts in unit time as a proxy of contribution and the number of customer comments in unit time as a proxy of creation. We also used aggregation method to calculated the average of multiple indicator variables. Tenure of membership was measured by calculating the mean of all customers' membership tenure in the live streaming room, and popularity was measured as the number of concurrent customers of the live streaming room. Table 1 shows all the variables and measurements.

5. Results

5.1. The empirical model

We estimate two models, that is, H1a-H3b (models for tie strength) and H4a, H4b, H5 (models for customer engagement behavior).

Table 2 covers the results of data analysis for model testing of H1a–H3b. The fixed effects within this model are included to capture time-independent heterogeneity. Model 0 includes the responsiveness and its square term (H1a). The beta parameter of the responsiveness is positive and significant ($p < .01$), and the parameter of the square term of the responsiveness is negative and significant ($p < .1$) as Model 0 shows, implying that the responsiveness has an inverted U-shaped relationship with tie strength, which supports H1a. The inverted U-shaped relationship is plotted in Fig. 3. The plot shows that the sweet spot in the responsiveness is around 0.190, which means that when sellers respond to approximately one-fifth of customers' questions, they are more effective in strengthening their ties with customers.

Model 1 includes personalization and its square term (H1b). The beta parameter of personalization is positive and significant ($p < .01$), and the parameter of the square term of the personalization is negative and significant ($p < .1$), as shown in Model 1, indicating that personalization has an inverted U-shaped relationship with tie strength, which supports H1b. The inverted U-shaped relationship is plotted in Fig. 4. Through

Table 1
Variables and measurements.

Variables	Measurements	Proxies	References
Responsiveness	Intensity of interactivity	Response rate for customers' questions in unit time	Lee (2005)
Personalization	Richness of interactivity	Average length of response content in unit time	Lee (2005) and Liu et al. (2019)
Tie strength	Volume of the breadth and depth of communication	The sum of comments which are talking about broadcasters in unit time and the sum of comments talking about products in unit time	Risselada et al. (2014)
Customer engagement behaviors	Consumption, contribution and creation	The mean of the number of liking in unit time, the number of gifts in unit time and the number of customer comments in unit time	Schivinski et al. (2016) and Liu et al. (2019)
Tenure of memberships	Customer familiarity	The mean of all customers' membership tenure in the live streaming room	Harris et al. (2016)
Popularity	Service quality	The number of concurrent customers of the live streaming room	Caminal and Vives (1996)

responsiveness and personalization, we find that interactivity has an inverted U-shaped relationship with tie strength.

Models 2 and 3 prove the moderating outcomes between responsiveness and tie strength. Model 2 examines the moderating effect of tenure of membership. Responsiveness and tenure of membership have a positive and significant interaction effect ($p < .05$). The interaction between tenure of membership and responsiveness squared is negative and significant ($p < .01$). The explanation of these parameters is that tenure of membership magnifies the inverted U-shaped relationship between responsiveness and tie strength, supporting H2a. Fig. 5 plots the interaction. Model 3 presents the interactions of responsiveness and popularity (H3a). The parameter of the interaction term between responsiveness and popularity is positive and significant ($p < .1$). The parameter of the interaction term between responsiveness squared and popularity is negative and significant ($p < .1$). The outcomes indicate that popularity increase the positive effect of the response rate on low tie strength and the negative effect of the response rate on a high level of tie strength. The interaction effect is plotted in Fig. 6. The plot proves that high popularity weakens the inverted U-shaped relationship between responsiveness and tie strength, which supports H3a. Note that with an improvement in popularity, the sweet spot decreases, indicating that high popularity in live streaming room encourages more customers with weak ties to participate the live streaming community.

Models 4 and 5 examine the moderating effects between personalization and tie strength. Model 4 shows the interaction of personalization and tenure of membership (H2b). Personalization and tenure of membership have a positive interaction impact, which is significant ($p < .1$). The interaction between tenure of membership and personalization squared is negative significantly ($p < .1$). The interpretation of these parameters is that tenure of memberships enhances the inverted U-shaped relationship between personalization and tie strength, supporting H2b. Fig. 7 plots the interaction effect. Model 5 demonstrates the interaction of personalization and popularity (H3b). In addition, personalization and popularity have a positive interaction effect, which is significant ($p < .05$). The interaction between customer ratings and personalization squared is negative and significant ($p < .1$). The explanation for these parameters is that popularity weakens the inverted U-shaped relationship between personalization and tie strength, which

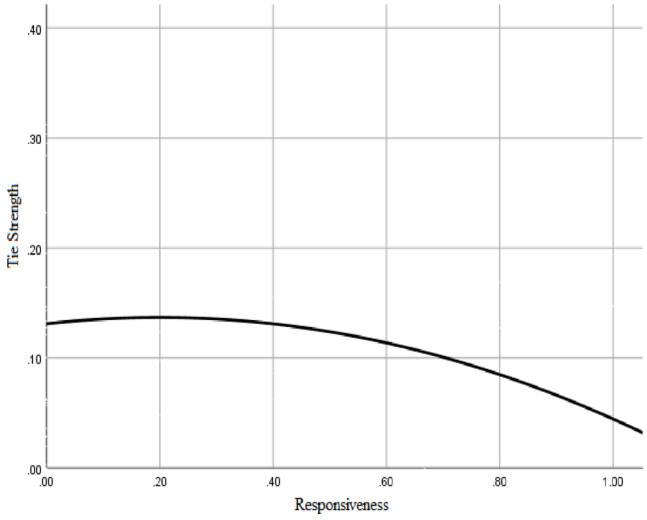


Fig. 3. Tie strength across the range of responsiveness.

Table 2
Effects models for tie strength.

	Model 0		Model 1		Model 2		Model 3		Model 4		Model 5	
Predictors												
Responsiveness	0.02**	(0.00)			0.00*	(0.00)	0.01	(0.00)				
Responsiveness squared (R2)	−0.01+	(0.00)			−0.02+	(0.01)	−0.00*	(0.00)				
Personalisation			0.00**	(0.00)					0.00*	(0.03)	0.02+	(0.05)
Personalisation squared (P2)			−0.01+	(0.00)					−0.00**	(0.01)	−0.01+	(0.00)
Interactions												
Responsiveness × Tenure of memberships					0.00*	(0.02)						
R2 × Tenure of memberships					−0.04**	(0.01)						
Responsiveness × Popularity							0.00+	(0.00)				
R2 × Popularity							−0.02+	(0.03)				
Personalisation × Tenure of memberships									0.01+	(0.00)		
P2 × Tenure of memberships									−0.01+	(0.00)		
Personalisation × Popularity											0.00*	(0.00)
P2 × Popularity											−0.01+	(0.00)

Robust standard errors are in parentheses.
+ $p < 0.1$.
* $p < 0.05$.
** $p < 0.01$.*** $p < 0.001$.

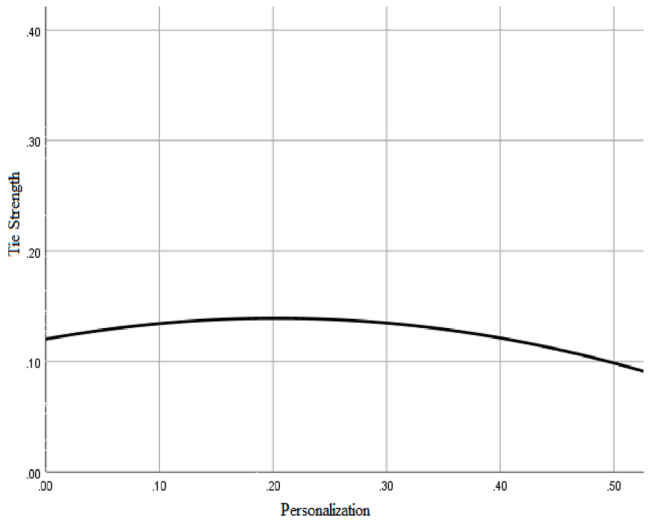


Fig. 4. Tie strength across the range of personalisation.

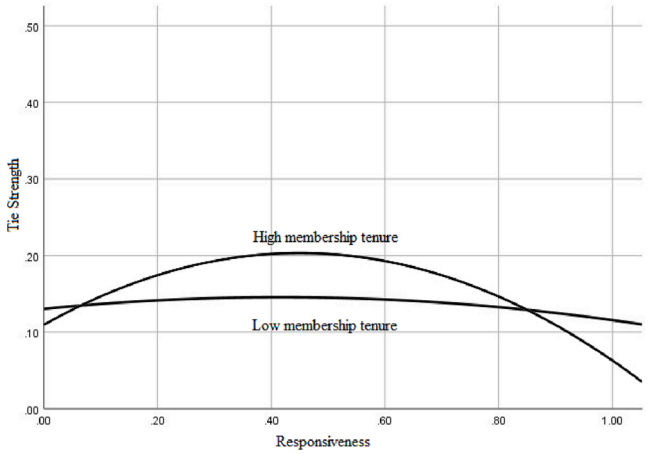


Fig. 5. Tenure memberships between responsiveness and tie strength.

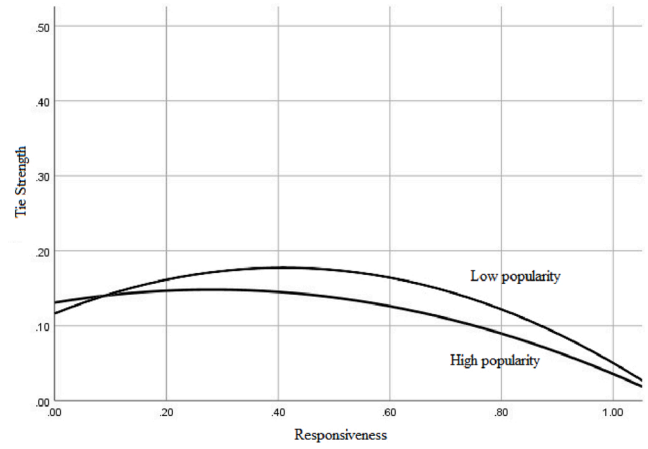


Fig. 6. Popularity between responsiveness and tie strength.

supports H3b. The interaction is plotted in Fig. 8.

Table 3 shows the results of testing for the relationship between interactivity and customer engagement behavior. Model 7 shows the interaction of responsiveness and customer engagement behaviors (H4a). Responsiveness and customer engagement behaviors have a

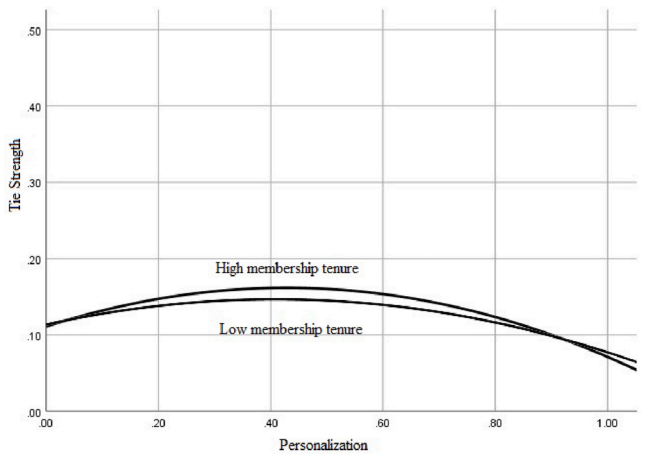


Fig. 7. Tenure memberships between personalization and tie strength.

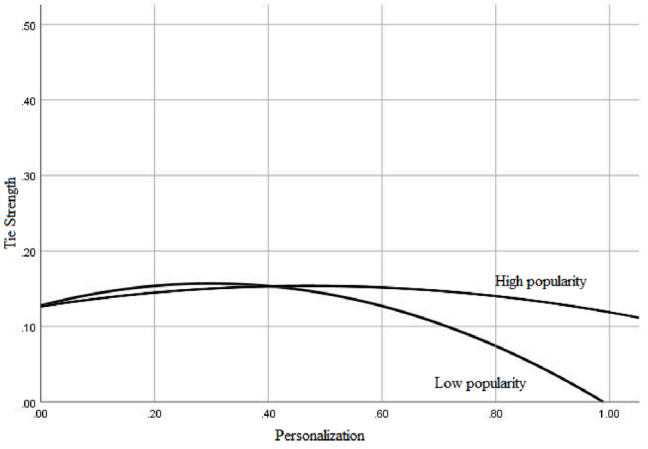


Fig. 8. Popularity between personalization and tie strength.

positive interaction effect, which is significant ($p < .01$). The interaction between responsiveness squared and customer engagement is significantly negative ($p < .05$). However, Model 8 shows that when both responsiveness and tie strength are considered, responsiveness becomes insignificant ($p > .05$) while tie strength has a significant effect on customer engagement behavior ($p < .01$). We plotted the relationship in Fig. 9. Model 9 shows the interaction of personalization and customer engagement behavior (H4b). Personalization and customer engagement behavior have a significantly positive interaction effect ($p < .1$). The interaction between personalization squared and customer engagement behavior is negative and significant ($p < .05$). Model 10 shows that when both personalization and tie strength are considered, personalization becomes insignificant ($p > .05$) while tie strength has a significant effect on customer engagement behavior ($p < .001$). We plotted the relationship in Fig. 10. Besides, the results reveal that tie strength plays a complete mediating role between interactivity and customer engagement behavior, thus supporting H6.

Table 3 also shows the results of testing for the relationship between tie strength and customer engagement behavior. In Model 6, we add the main effect of tie strength (H5). The measures prove that tie strength positively effects on customer engagement behavior ($p < .01$). The beta parameter of the squared term of tie strength is also positive and significant ($p < .1$). The results reveal that tie strength has a positive relationship with customer engagement behavior. Consequently, H5 is supported. The relationship is plotted in Fig. 11, where it shows that with an increase in tie strength, the degree of customer engagement behavior will increase correspondingly, but the rate of improvement

Table 3
Effects models for customer engagement behavior.

	Model 6		Model 7		Model 8		Model 9		Model 10	
Predictors										
Responsiveness			0.02**	(0.00)	0.03	(0.01)				
Responsiveness squared(R2)			−0.01*	(0.01)	0.01+	(0.02)				
Personalisation							0.00+	(0.00)	0.02	(0.01)
Personalisation squared(P2)							−0.03*	(0.00)	−0.01	(0.00)
Tie strength	0.01**	(0.00)			0.02**	(0.00)			0.00***	(0.02)
Tie strength squared	0.00+	(0.02)			0.04*	(0.01)			0.01**	(0.01)

Robust standard errors are in parentheses.
+ p 0.1.
* $p < 0.05$.
** $p < 0.01$.
*** $p < 0.001$.

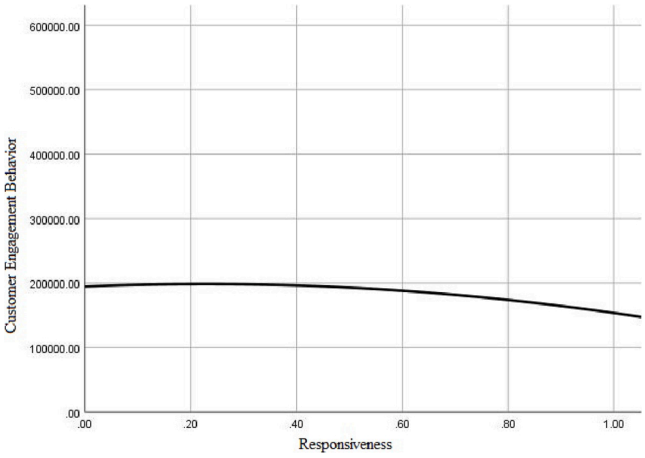


Fig. 9. Customer engagement across the range of responsiveness.

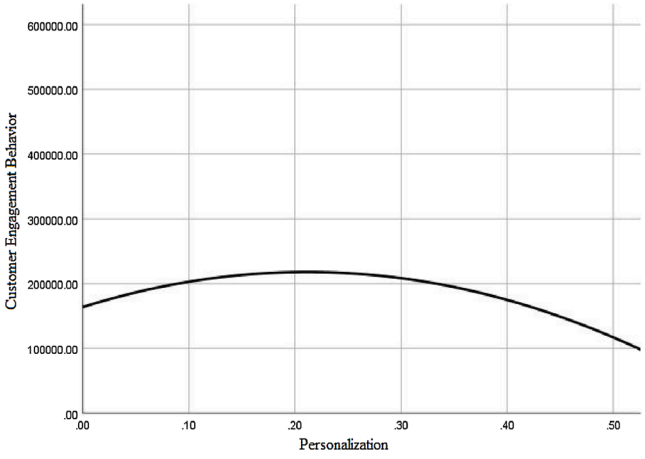


Fig. 10. Customer engagement across the range of personalization.

gradually slows.

Moreover, we conducted additional analyses to examine the robustness of the results of the empirical tests. We used generalized estimating equations for the data analysis to produce more effective regression parameters (Prentice, 1986). The results generated from all robust analyses remain unchanged.

6. Discussion

This study examines the dynamic effect of interactivity on customer engagement behaviors through tie strength in the context of live

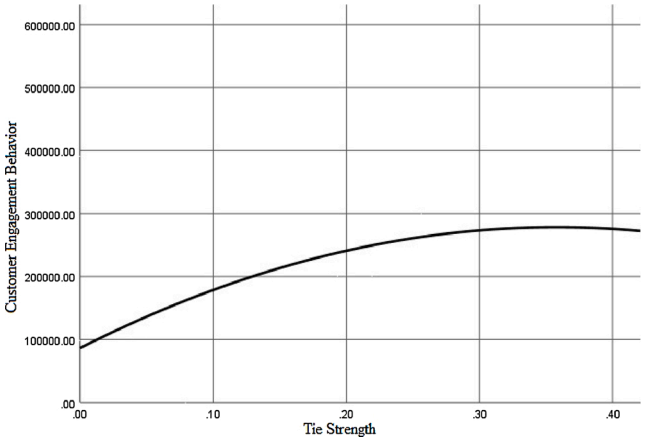


Fig. 11. Customer engagement across the range of tie strength.

streaming commerce. Drawing on the S-O-R framework, we constructed a set of variables from real-time online reviews using text mining techniques. An empirical study was conducted using the data collected from crawling one of the largest live streaming commerce platforms in China. We found several interesting results. Firstly, the results support that interactivity as a social stimulus reported a nonlinear effect on customer engagement behaviors in live streaming commerce platforms. More precisely, increasing the interaction between customers and broadcasters or among customers will promote customers to thumb up and send gifts actively. However, when the interaction exceeds a certain point, it will lead to customer avoidance. These findings reconcile the inconsistencies in prior literature concerning the influence of interactivity on customer engagement (Nitzan & Libai, 2011; Ul Islam & Rahman, 2017; Vries et al., 2012). Receiving such finding indicates that interactivity, as an important feature of live streaming commerce, will be both beneficial and troublesome for customers.

To explore the relationship between interactivity and customer engagement behavior, we divide interactivity into responsiveness and personalization. On the one hand, the findings reveal that responsiveness has a nonlinear (inverted U-shaped) relationship with customer engagement behavior. Hence, it not only supports the findings of Tsai and Kang (2019), who noted that information exchanged by responding to customer's needs arouses their behavior intentions in the online community. It also supports the results of (Von Hippel, 1994) that responding to several customers' fake demands may cause customers' reactance and avoidance. On the other hand, personalization was found to have a nonlinear relationship with customer engagement behavior. When compared to responsiveness, customer engagement behaviors are more easily affected by personalization. This was explained that when compared with the number or frequency of interactions, the content of interactions arouses customer reactions more intensively.

Next, tie strength as an organism between interactivity and customer engagement behaviors plays a key role in this study. The findings reveal that interactivity has an inverted U-shaped influence on the strength of social ties in live streaming commerce, which means interactivity does not always have a positive impact on tie strength. When interactivity exceeds a certain threshold, it hinders the formation of close community relationships. Therefore, the findings of past studies by Mano and Mesch (2010) and Shen et al. (2010) are supported. Besides, we found that responsiveness has an inverted U-shaped relationship with tie strength. This result reconciles the findings of Zhao and Lu (2012) and Zhou et al. (2018). According to the Social Exchange Theory, Zhao and Lu (2012) pointed that responding to customers' needs can increase customer satisfaction and trust, promoting close social relations, while Zhou et al. (2018) found over-responding to customers' demands may cause learning myopia and hinder the product or service developers in attracting new customers. Similarly, we also found personalization has an inverted U-shaped relationship with tie strength. This finding, consistent with Van Doorn and Hoekstra (2013), indicates that personalized communication may enhance customers' positive attitudes towards the source, but using personal information may induce feelings of intrusiveness, which will hamper the building of close community relationships.

Regarding tie strength and customer engagement behavior in live streaming commerce, our result is consistent with Xu et al. (2012), where tie strength has a positive relationship with customer engagement behavior. This result indicates that when the community relationship changes from weak to strong, customers' willingness to thumb up, send e-WOM and product referral gradually becomes stronger. Surprisingly, we found that the strength of social ties acted as a complete mediator between interactivity and customer engagement behavior. This result proves the importance of creating harmonious community relationships (Luo et al., 2016).

Finally, we verified the role of two moderating variables, tenure of membership and popularity. As expected, tenure of membership significantly strengthens the inverted U-shaped relationship between interactivity and tie strength. This finding is in line with Luhmann et al. (1979). Tenure of membership may induce either strong or weak ties, depending on whether the customer's attitude (induced by interactivity) towards the company is positive or negative in live streaming commerce. In particular, we found that tenure of membership strengthens the inverted U-shaped relationship between responsiveness and tie strength. It confirms the findings of past studies by Sreejesh et al. (2020) and Fred et al. (2020). This finding indicates that experienced customers are more sensitive to the changes in responsiveness than new ones. Similarly, tenure of memberships amplifies the inverted U-shaped relationship between personalization and tie strength. These findings add values to the existing literature, especially in the context of live streaming commerce.

One of the most interesting findings of this study is the moderating roles of popularity. The findings reveal that popularity mitigates the inverted U-shaped relationship between interactivity and tie strength. This implies that popularity alleviates both the positive and negative effects of interactivity on tie strength in live streaming commerce context. This explains that popularity not only represents perceived value but also can lead to herd behaviors (Banerjee, 1992; Pavlou & Gefen, 2004; Wan et al., 2018; Zhu & Zhang, 2010). It is a significant finding that may prompt broadcasters to use popularity to mitigate the negative impact of interactivity on tie strength. As expected, we found that popularity not only weakens the inverted U-shaped relationship between responsiveness and tie strength but also mitigates the inverted U-shaped relationship between personalization and tie strength. These findings are explained that responsiveness and personalization are both considered as important indicators of service quality (Parasuraman et al., 2005).

6.1. Theoretical contributions

Our research has several theoretical implications. First of all, we contribute to the literature by investigating the interactivity of the live streaming community. Previous literature has mainly focused on the customer perspective to investigate customer motivation or perceived value (Apiradee & Nuttapol, 2018; Cai et al., 2018), while this study considers the technical features of the live streaming commerce platforms, and the biggest feature is interactivity between customers and broadcasters and among customers (Scheibe et al., 2016; Zhou et al., 2019). For interactivity, most studies have focused on online brand community (Yang, 2008). These virtual communities provide an environment where customers can communicate and share information with each other (Cyr et al., 2009). However, these communities mainly interact in the form of messages, which may lead to the hysteresis of information transmission. Social interaction theory states that interactivity reflects the degree of real-time and information exchange (Jang, Olfman, Ko, Koh, & Kim, 2008; Steuer, 2000). Exploring the effect of interactivity under inadequate real-time and information exchange may lead to variation and inaccurate results. Hence, investigating interactivity in live streaming communities can provide a more accurate understanding. Thus, this study fills the gap by focusing on interactivity in the context of live streaming commerce.

Second, we contribute to the literature on customer engagement behavior by measuring actual non-transactional behaviors in the context of live streaming commerce. Prior literature has conducted in-depth studies on customer engagement (Brodie, Hollebeek, et al., 2011; Brodie, Juric, et al., 2011; Van Doorn et al., 2010), but there is an inconsistency in the conceptualization and operationalization of customer engagement in these studies. For example, although Jaakkola and Alexander (2014) defined customer engagement as a multi-dimensional structure including cognition, emotion, and behavior, due to limited technology, they only measured the behavioral dimension. Similarly, Van Doorn et al. (2010) defined customer engagement as a non-transactional behavior, yet measured only behavioral intentions instead of actual behaviors. Moreover, Brodie, Juric, et al. (2011) pointed out that customer engagement behavior has a dynamic nature, but in the actual measurement, they assumed that the process is constant. Therefore, this study fills this gap by analyzing customers' actual dynamic behaviors in live streaming commerce platforms (e.g. sending gifts, liking, and commenting), which are the indicators for measurement of customer engagement behavior. We collected big data from one of the most popular live streaming commerce platform in China, and this large sample supports the representativeness of our findings, while avoiding the subjectivity and time-consuming nature of questionnaires (Kar & Dwivedi, 2020).

Third, this study also contributes a novel research model that is structured according to the S-O-R model and establishes a theoretically grounded link between interactivity and customer engagement behavior through the motivation of building close community relationships in the context of live streaming commerce. While previous research findings have indicated the effect of interactivity on customer engagement behavior, no consensus has been reached so far. Ul Islam and Rahman (2017) explained that the effect of interactivity on customer engagement behavior is positive, while social interactions may cause customer defection behavior in Nitzan and Libai (2011)'s perspective. Thus, this study analyses the comprehensive influence of interactivity on customer engagement behavior from a dynamic perspective in the live streaming commerce platform and finds that the relationship is inverted U-shaped. Besides, prior literature has not thoroughly studied the organism of interactivity on customer engagement behaviors, though the interactivity of live streaming community generates customer non-transactional behaviors by influencing social relationships (Luo et al., 2016). Therefore, this paper fills this gap by studying the effect of tie strength on interactivity and customer engagement behavior, and finds that tie strength plays the role of a mediator.

6.2. Implications for practice

The study model and findings have several important implications for broadcasters or sellers wishing to establish close relationships with customers and improve customer engagement behavior. First, in live streaming commerce, broadcasters should encourage the interactions between customers which will affect their engagement behavior in the live streaming room, such as gifting and commenting. However, the spontaneous behaviors of customers keep developing and changing. Hence, broadcasters should take measures to find the inflection point that affects the behaviors. When customers are engaging more, broadcasters should strengthen the interactivity in the live streaming room. Specifically, they need to pay attention to both interaction frequency and interaction content, which may further promote the engagement of customers. When engaging behavior in recession, broadcasters should take measures to slow down the process of recession. If there is a negative eWOM in the live streaming room, which may cause customer dissatisfaction, broadcasters should find ways to change the customer's attitude, such as by offering hair coupons, introducing new products, or through social media propagation force to promote personal charm, etc.

Second, as an opinion leader and information source, broadcasters should give importance to developing close relationships with customers during the live streaming process, because the strength of social ties positively affects customer engagement behavior. When the atmosphere of live streaming is harmonious and the forms of interaction are abundant, customers will consciously surround themselves with the broadcasters and form a close relationship with them. Also, customers' familiarity with live streaming commerce platforms will affect the formation of close community relations. Therefore, for new customers, before live streaming, broadcasters can publicize through social media, such as Sina Weibo, in order to make them familiar with the platforms. For regular customers, they should be further given coupons to encourage participation in the live streaming community. However, when too much information or low-quality information (expletives, meaningless emotions) fills the live streaming room, it may lead to customer dissatisfaction and damage community relations. Therefore, broadcasters should manage the live streaming room and promptly silence users whom post swear words or meaningless emotions. They can subdivide customers using big data technology and carry out product communication based on different user portraits. Moreover, broadcasters can also invite celebrities or experts to increase the popularity of the live streaming room, and slow down the destruction of community relationships.

Third, broadcasters should pay attention to the importance of building a harmonious and close virtual community. Some scholars have pointed out that it is easy to apply for live streaming on live commerce platforms, but it is not easy to form a harmonious community with customers (Cai et al., 2018). Therefore, broadcasters should strengthen their professional quality and learn from the construction process of the brand community and the knowledge-sharing community. With the help of interactive electronic platforms, they could create greater value for customers.

6.3. Limitations and future directions

A few limitations of our study elucidate further directions for future studies. First, our study tracked real-time online data from customers in live streaming communities who had experience watching Weibo Live for marketing purposes. However, in future studies, the research scope of our study could be extended to other platforms (such as Instagram, Facebook, Twitter, and Taobao), other product categories (electronic products, cosmetics, etc.), and other countries, particularly in Western contexts. Second, in our study, online reviews are categorized with attributes instead of emotions. In fact, customer attitudes also have a significant impact on customer demand, so future research should study how customer attitudes affect demand through text emotion analysis. Third, the impact of spam reviews is excluded in our research. In fact, a small number of customers post emojis or other useless symbols in live streaming, which influences the research results to some extent. Therefore, an analysis of this practice is helpful. However, identifying fake reviews is always a challenging work (Li, Chen, Mukherjee, Liu, & Shao, 2015; Sandulescu & Ester, 2016). Depending on the development of future methods for detecting spam reviews, an extension of our research could assess the influence of spam comments on the results of our study.

7. Conclusions

Triggered by the exponential rise in live streaming commerce, this study attempts to gain a better understanding of interactivity and customer engagement behavior in the field. The S-O-R model was employed as the theoretical framework to identify the effect of interactivity on customer engagement through tie strength. Based on real-time online reviews clawed from one of the biggest live streaming commerce platforms in China, the study confirmed that interactivity will have an inverted U-shaped influence on customer engagement behavior. Moreover, we surprisingly found that the strength of social ties act as a complete mediator between interactivity and customer engagement behavior. We also verified the role of two moderating variables, membership tenure and popularity. In light of the rich findings and ensuing discussions of this research, both academicians and practitioners would be interested to explore the different ways to enhance customer engagement in the future.

Author statement

Kai Kang: conceptualization, methodology, resources, writing – original draft. Jinxuan Lu: writing – original draft, software, validation, investigation, resources, data curation. Lingyun Guo: supervision, project administration, formal analysis, conceptualization. Wenlu Li: writing – review & editing.

Conflict of interest

No conflict of interest.

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